EASY

PROBLEM-SOLVING AND DATA ANALYSIS

Two-variable data: Models and scatterplots

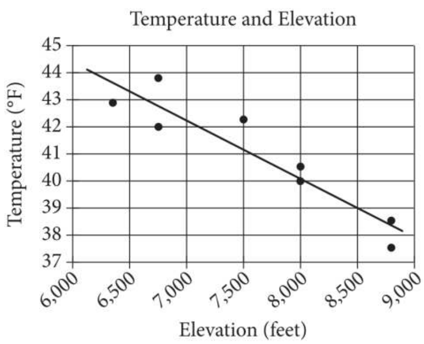
02

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 PROBLEM-SOLVING AND DATA ANALYSIS



The scatterplot above shows the high temperature on a certain day and the elevation of 8 different locations in the Lake Tahoe Basin. A line of best fit for the data is also shown. Which of the following statements best describes the association between the elevation and the temperature of locations in the Lake Tahoe Basin?

A. As the elevation increases, the temperature tends to increase.

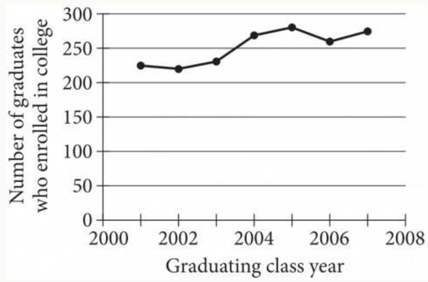
B. As the elevation increases, the temperature tends to decrease.

C. As the elevation decreases, the temperature tends to decrease.

D. There is no association between the elevation and the temperature.

Q2

PROBLEM-SOLVING AND DATA ANALYSIS



The line graph shows the number of graduates from the classes of 2001 through 2007 at a certain school who enrolled in college within 24 months of graduation. Of the following, which class had the fewest graduates who enrolled in college within 24 months of graduation?

A. 2002

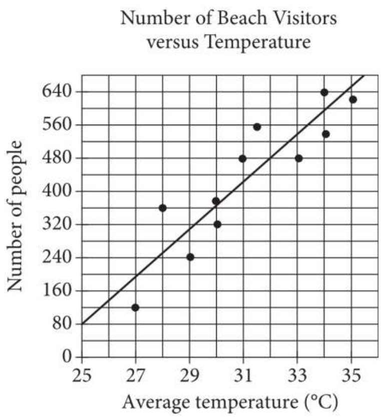
B. 2004

C. 2005

D. 2007

Q3

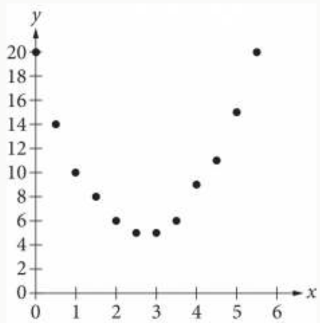
GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS



Each dot in the scatterplot above represents the temperature and the number of people who visited a beach in Lagos, Nigeria, on one of eleven different days. The line of best fit for the data is also shown. According to the line of best fit, what is the number of people, rounded to the nearest 10, predicted to visit this beach on a day with an average temperature of 32°C?

STUDENT-PRODUCED RESPONSE

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



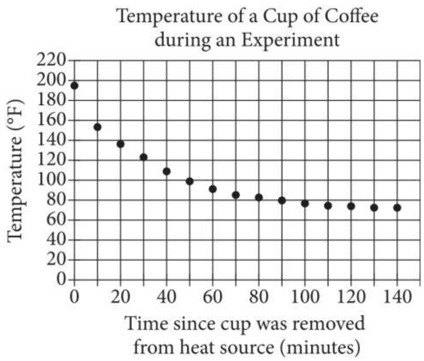
Of the following, which is the best model for the data in the scatterplot?

A. $y = 2x^2 - 11x - 20$

B. $y = 2x^2 - 11x + 20$

C. $y = 2x^2 - 5x - 3$

D. $y = 2x^2 - 5x + 3$



In an experiment, a heated cup of coffee is removed from a heat source, and the cup of coffee is then left in a room that is kept at a constant temperature. The graph above shows the temperature, in degrees Fahrenheit (°F), of the coffee immediately after being removed from the heat source and at 10-minute intervals thereafter. During which of the following 10-minute intervals does the temperature of the coffee decrease at the greatest average rate?

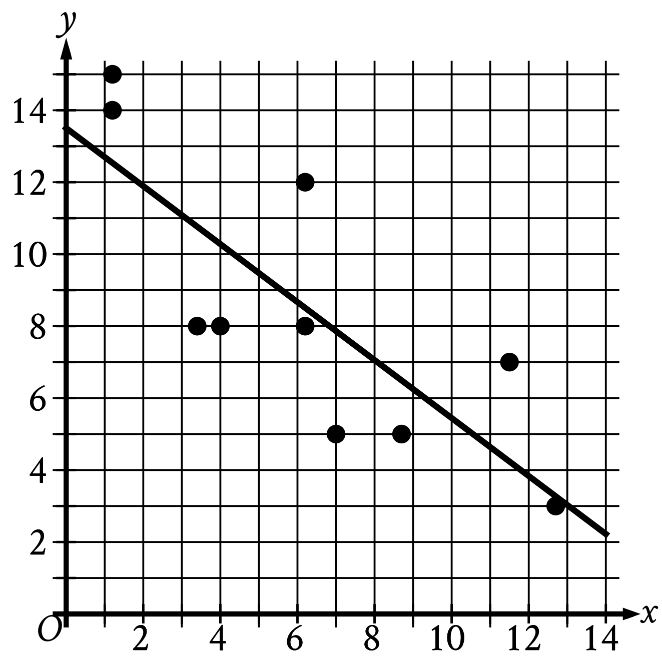
A. Between 0 and 10 minutes

B. Between 30 and 40 minutes

C. Between 50 and 60 minutes

D. Between 90 and 100 minutes

The scatterplot shows the relationship between two variables, x and y. A line of best fit is also shown.



- The scatterplot has 10 data points.
- The data points are in a linear pattern trending down from left to right.
- A line of best fit is shown:
  - The line of best slants down from left to right.
  - 1 point is touching the line of best fit.
  - 4 points are above the line of best fit.
  - 5 points are below the line of best fit.
  - The line of best fit passes through the following approximate coordinates:
    - (2 comma 12)
    - (8 comma 7)
    - (13 comma 3)

Which of the following equations best represents the line of best fit shown?

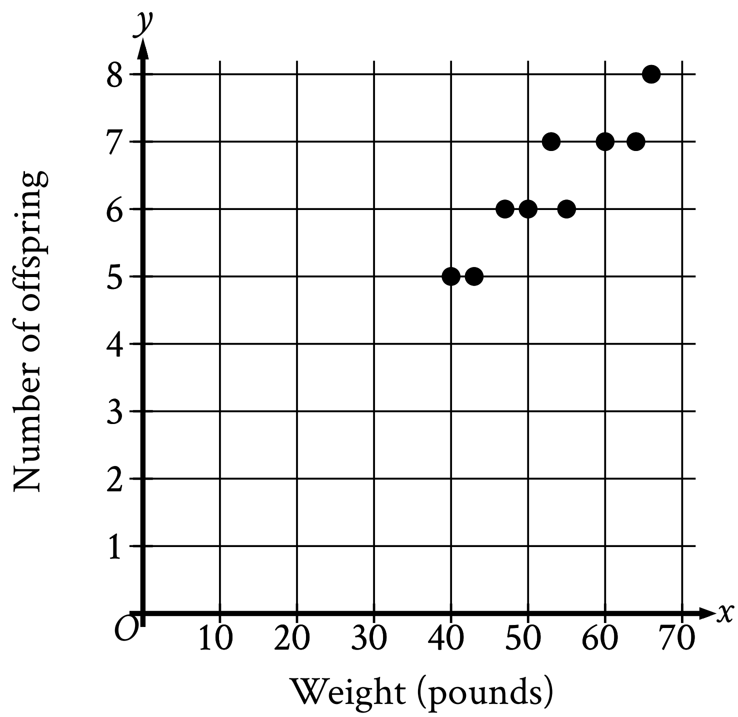
A. $y = 13.5 + 0.8x$

B. $y = 13.5 - 0.8x$

C. $y = -13.5 + 0.8x$

D. $y = -13.5 - 0.8x$

The scatterplot shows the relationship between the weight, in pounds, of each of 9 female gray wolves on April 30 and the number of offspring each gray wolf produced.



- The scatterplot has 9 data points.
- The data points are in a linear pattern trending up from left to right.
- The data points have the following coordinates:
  - (40 comma 5)
  - (43 comma 5)
  - (47 comma 6)
  - (50 comma 6)
  - (53 comma 7)
  - (55 comma 6)
  - (60 comma 7)
  - (64 comma 7)
  - (66 comma 8)

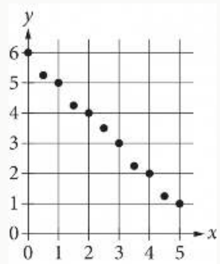
How many offspring did the 50-pound gray wolf produce?

A. 8

B. 7

C. 6

D. 5



Which of the following could be an equation for a line of best fit for the data in the scatterplot?

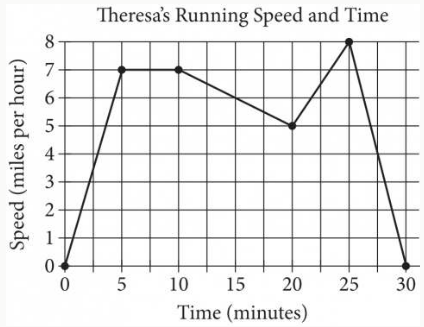
A. $y = -x + 6$

B. $y = -x - 6$

C. $y = 6x + 1$

D. $y = 6x - 1$

Q9 PROBLEM-SOLVING AND DATA ANALYSIS



Theresa ran on a treadmill for thirty minutes, and her time and speed are shown on the graph above. According to the graph, which of the following statements is NOT true concerning Theresa's run?

A. Theresa ran at a constant speed for five minutes.

B. Theresa's speed was increasing for a longer period of time than it was decreasing.

C. Theresa's speed decreased at a constant rate during the last five minutes.

D. Theresa's speed reached its maximum during the last ten minutes.

Q10 PROBLEM-SOLVING AND DATA ANALYSIS

The table shows selected values from function f.

x	$f x$
-1	16
0	17
1	18
2	19

Which of the following is the best description of function f?

A. Decreasing linear

B. Increasing linear

C. Decreasing exponential

D. Increasing exponential